

FATEMEH MIRZAABOLHASSANI

Computational Biology | Single-Cell Cancer Genomics

✉ Fateme.abolhassani91@gmail.com 📍 Tehran, Iran 🌐 GitHub 🌐 Fateme Abolhassani 📁 Portfolio

FIELDS OF INTEREST

- Cancer Cell States & Heterogeneity
- Gene Regulatory Networks
- Single-Cell & Spatial Transcriptomics
- Tumour-Immune Interactions
- Interpretable Modeling for Precision Oncology

RESEARCH EXPERIENCE

Bioinformatics Research Collaborator — Thyroid Cancer

📅 Sept. 2024 – present 📍 EMRI, Tehran University of Medical Sciences

- Led computational analyses of single-cell RNA-seq data from normal thyroid, papillary thyroid carcinoma (PTC), and anaplastic thyroid carcinoma (ATC), from quality control and cell annotation through biological interpretation.
- Integrated pseudotime-dependent expression, gene regulatory networks, and inferred copy-number alterations to prioritize transcription factors associated with progression, dedifferentiation, and loss of thyroid epithelial identity.
- Collaborated with clinicians and experimental biologists to prioritize candidates for validation and currently contributes to manuscript interpretation, figures, and writing.
- Supervisors: Dr. Kaveh Kavousi and Dr. Vahid Haghpanah.

M.Sc. Thesis and Continuing Research — Glioblastoma

📅 2022 – 2023 & 2025 – present 📍 Sharif University of Technology

- Developed a framework comparing transcriptional pseudotime with phylogenetic relationships inferred from RNA-derived mutation information to prioritize progression-associated genes in glioblastoma.
- Completed the M.Sc. thesis in 2023 and presented the work in 2025 📁 Repo; the current extension investigates joint RNA-mutation representations and the reliability of integrating sparse molecular signals.
- Affiliations: 🌐 CBB Lab, University of Tehran, 🌐 CBRC Lab, Amirkabir University of Technology.
- Supervisors: Dr. Mohammad Hadi Foroughmand Araabi, Dr. Kaveh Kavousi, and Dr. Fatemeh Zare-Mirakabad.

TECHNICAL SKILLS

COMPUTATIONAL FOUNDATIONS

Python · R · Bash/Linux · Git · Docker · HPC

SINGLE-CELL TRANSCRIPTOMICS

STAR · Seurat · Monocle3 · tradeSeq · cNMF

Read alignment & preprocessing · QC/annotation · integration · trajectory/pseudotime · expression dynamics · transcriptional programs

CANCER GENOMICS & REGULATION

pySCENIC · CellOracle · CopyKAT · inferCNV

GRN inference · CNV inference · RNA-derived SNV analysis · mutation phylogenetics · RNA-mutation/WNN integration

STATISTICAL & INTEGRATIVE ANALYSIS

Differential & pseudobulk analysis · pathway analysis · bulk RNA-seq analysis · multimodal/evidence integration

RESEARCH PRACTICE

Biological question → computational strategy · interdisciplinary work with clinicians and experimental biologists · method evaluation & robustness · biological interpretation → testable hypotheses

ADDITIONAL METHODS

Molecular docking & MD · GROMACS · AutoDock Vina

RESEARCH OUTPUTS

Mirzaabolhassani F., et al. “Integrative Single-Cell Transcriptomics to Identify Regulatory Drivers in the Dedifferentiation of Thyroid Cancer.” *Manuscript in preparation. Co-first author.*

Identification of Driver Genes in Glioblastoma Based on Single-Cell Gene Expression Data Using Integrated Pseudotime and Phylogenetic Analysis

- **Poster**, Iran Conference on Bioinformatics, Zanjan, Iran, Feb. 2025. [🔗 Abstract \(p. 140\)](#) [🔗 Poster](#)
- **Oral presentation**, Computational Biology Research Center, Amirkabir University of Technology, Tehran, Iran, Apr. 2025.

ADDITIONAL RESEARCH EXPERIENCE

Exploratory Doxorubicin–DNA Docking and Molecular Dynamics

- [📅 2025](#) [📍 Interdisciplinary research collaboration](#)
- Performed molecular docking and initial molecular-dynamics simulations of doxorubicin–DNA binding using AutoDock Vina and GROMACS in collaboration with researchers in biology and computational physics.

RNA Secondary-Structure Prediction

- [📅 Feb. 2024 – May 2024](#) [📍 Amirkabir University of Technology](#)
- Implemented methods combining stochastic context-free grammars and evolutionary history; supervised a B.Sc. student. Supervisor: [Dr. Fatemeh Zare-Mirakabad](#). [🔗 Repo](#)

TEACHING & ACADEMIC SERVICE

Teaching Assistant

- [📅 2018 – 2023](#) [📍 Sharif, Amirkabir, and Alzahra Universities](#)
- Foundations of Computation Theory, Databases, and Data Structures and Algorithms.

Next-Generation Sequencing Workshop Instructor

- [📅 Sept. 2024](#) [📍 Pasteur Institute of Iran, Tehran](#)
- Delivered Ph.D.-level training on next-generation sequencing techniques.

ICPC Team Coach — Asia Tehran Regional Contest

- [📅 2025](#) [📍 Tehran, Iran](#)
- Mentored a competitive programming team in problem-solving, strategy, and contest preparation.

Co-founder, Computer Science Journal Club

- [📅 2018 – 2020](#) [📍 Alzahra University](#)

SELECTED HONORS

- Admitted to the M.Sc. programs at Sharif University of Technology, University of Tehran, and Amirkabir University of Technology through the exceptionally talented student scheme, with exemption from the national entrance examination.
- Received governmental fellowships for outstanding Bachelor performance.
- Admitted to NODET (National Organization for Development of Exceptional Talent) via competitive entrance exam for high school.

REFERENCES

Dr. Kaveh Kavousi

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Dr. Fatemeh Zare-Mirakabad

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LANGUAGES

English

TOEFL iBT: 4/6